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A flexible cost model for seaport-hinterland decisions in container shipping

Thiago de Almeida Rodrigues [a,*], Caroline Maria de Miranda Mota [b], Udechukwu Ojiako [c,d], Maxwell Chipulu [e], Alasdair Marshall [f], Fikri Dweiri [g]

[a] Universidade Federal do Espírito Santo, Vitória, Espírito Santo, Brazil
[b] Universidade Federal de Pernambuco, Recife, Pernambuco, Brazil
[c] University of Strathclyde, UK
[d] University of Hull, UK
[e] Edinburgh Napier University, UK
[f] University of Southampton, UK
[g] University of Sharjah, United Arab Emirates

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ABSTRACT

A seaport–hinterland system offers a wide range of service options for shippers and consignees to export or import containerized cargo. The present study, set within the context of Brazil, seeks to propose a flexible cost model to aid joint seaport–hinterland network decisions by freight forwarders and clients. More specifically, employing simulation and sensibility analysis, the proposed model is employed to explore: (i) decisions on whether to perform logistics services in seaports or dry ports; (ii) decisions on whether to transport containerized cargo by road or by multimodal transportation (road and rail); and (iii) decisions on whether cargo is best delivered or received in either consolidated or deconsolidated form. Simulations are undertaken against seven transportation strategy permutations via which service operators might deliver containers from seaports or dry ports to clients. We find five factors—(i) cargo value, (ii) cargo accommodation type, (iii) transportation mode, (iv) delivery distance and (v) shipping process time—as critical to joint seaport–hinterland operations impacting shippers and consignees. Furthermore, dry ports are shown to be particularly cost efficient for imports requiring long-term storage, even where the dry ports are not directly connected to seaports or intermodal facilities.

1. Introduction

An ever-increasing volume of international trade and maritime cargo transportation (and associated growth in cargo volumes) has intensified the pressure on seaports (Wang, Meng, & Jia, 2019) with wide implications for their operations (Bentaleb, Mabrouki, & Semma, 2015; Sarmadi, Amiri-Aref, Dong, & Hicks, 2020). Serving as the point of entry or exit for goods arriving or departing by sea, most seaports (especially those that are strategically located) are facing significant challenges relating to issues with terminal capacity required to handle ever larger-sized vessels. Seaports are also facing hinterland (i.e., cities, manufacturing centers, and distribution centers) access-related challenges (Khaslavskaya & Roso, 2020).

One of the main transport nodes in global international trade and maritime cargo transportation is the seaport container terminal. Located in seaports, seaport container terminals are specialized facilities typically comprising a terminal with a dock and a yard that are employed for the handling of several activities such as loading (also unloading) of shipping containers. Another important transport node is the dry port. As a form of inland intermodal terminal, dry ports are typically connected directly to seaports via high-capacity transport modes (e.g., railways) which allow clients to store and collect their containers as they might otherwise do from a seaport (Roso, Woxenius, & Lumsden, 2009). They are, therefore, a major transit facility used for storing and transferring containerized cargo. Their primary function within the logistic chain is to facilitate the handling and management of container flows and other connected inland transport modes (Fazi & Roodbergen, 2018) in an effective and efficient manner (Ha, Yang, Notteboom, Ng, & Heo, 2017).

The seaport–hinterland network, which describes the nature of the various nodes (points) within a seaport and its hinterland, provides a diverse set of options for services, spanning from the customer's facility

* Corresponding author.
E-mail addresses: thiago.a.rodrigues@ufes.br (T. de Almeida Rodrigues), caroline.mota@ufpe.br (C.M. de Miranda Mota), udechukwu.ojiako@outlook.com (U. Ojiako), m.chipulu@napier.ac.uk (M. Chipulu), a.marshall@soton.ac.uk (A. Marshall), fdweiri@sharjah.ac.ae (F. Dweiri).

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to the loading of containerized cargo onto vessels on the export side, and from the discharge of containers at the seaport yard to their transportation to the customer facility on the import side (Fazi & Roodbergen, 2018). They are, however, fraught with considerable complexity due to the nature of interconnectedness between the various nodes that are involved in seaport-hinterland container movement. Adding to this complexity is the reality that they vary considerably by functionality range and ownership structure (Khaslavskaya & Roso, 2020). This variability is inevitable, given that they often span multiple seaports and geographical regions (i.e., hinterlands). The logistic complexity of seaport–hinterland networks also stems from the myriad of road, rail, and inland waterways networks that connect seaports to hinterlands (Sdoukopoulos & Boile, 2020). Recognizing this complexity, the concept of the seaport–hinterland network is generally understood as referring to the entirety of logistics networks connecting seaports to inland regions, in some cases through dry ports (Wei, Sheng, & Lee, 2018). It has also been estimated that the costs of seaport–hinterland networks account for between 40% and 80% of total logistic costs (Haralambides, 2019; Notteboom & Rodrigue, 2005).

Drawing from the literature, there are two major determining viewpoints that appear to influence stakeholder decisions on whether to conduct their operations either through a dry port or through a seaport (Castelein, Geerlings, & van Duin, 2019). The first viewpoint suggests that clients (shippers/consignees) will select an ocean carrier and then delegate decisions relating to route and port choice to the ocean carrier in question, the implications being that clients (shippers/consignees) do not get involved in the final decision on choice of service provider (carrier haulage). Conversely, the second viewpoint is that decisions on the route and landside transportation, including the dry port and seaport choice, are made by the client (merchant haulage) (Castelein et al., 2019). Our study is framed within this viewpoint.

Regarding the clients that choose their services, providers can be grouped according to size and priorities. Thus, on size, smaller clients appear more likely to be influenced by price differentials while larger clients appear more likely to be influenced by service quality and reliability (Slack, 1985). On the other hand, as relates to priorities, clients can fall into three groups. (i) The first are those that maintain long-term contracts with shipping lines and are, therefore, more likely to be committed to a particular carrier. These clients are more likely to rely on the shipping lines' choice of transportation strategy. (ii) The second group comprises those that use freight forwarders to manage their import/export processes. These clients are more likely to delegate the responsibility for the transportation strategy to the freight forwarders. Finally, we have (iii) independent shippers/consignees that are usually engaged in defining their own transportation strategy to import/export a containerized cargo (Tongzon, 2009).

There are various reasons why cost considerations are an important stakeholder consideration in seaport–hinterland networks, particularly among service providers and clients (Rodrigues, Caroline Mota, and Inez Santos., 2021). First, costs are important because of their implications for competition (a key interest for service providers). For example, lower costs can enable the competitive pricing of goods and services by service providers which can attract and retain a greater number of clients (shippers/consignees), leading to increases in business volumes and a larger presence in the market. Second, cost efficiencies can reduce overall operational expenses (for the service provider), leading to greater profitability and the maximization of return on investments (ROI) across the entire seaport–hinterland network value chain.

Within this context, we opine that ensuring optimal (best) costs for seaport–hinterland decisions are likely to be dependent on developing close cooperation/collaboration between key seaport–hinterland stakeholders (Notteboom, 2008). We make this assertion based on two ideas. The first is that it is widely acknowledged that efficiencies and costs considerations/reduction are key drivers behind the decision of most firms and organizations to engage in collaborative relationships (Kim & Netessine, 2013). The second is the literature (e.g., Song & Panayides, 2008) which suggests that the nature of the relationship between key seaport–hinterland actors—in this instance, service providers and clients—serves a key driver for seaport–hinterland integration.

This should not be a surprise. In the literature, cooperation and collaboration have been consistently construed as ubiquitous in seaport–hinterland settings (see Arendt, 2020; Brooks, McCalla, Pallis, & Van De Lugt, 2010; Chen, Jeevan, & Cahoon, 2016; Jeevan et al., 2021; Kang & Kim, 2017; Nguyen, Thai, Nguyen, & Tran, 2021). Cooperation/collaboration among seaport–hinterland network stakeholders will allow for information asymmetry (Li, Fan, Lee, & Cheng, 2015), cognitive load (Kirschner, Sweller, Kirschner, & Jimmy, 2018), and opportunism reduction (Galvin, Tywoniak, & Sutherland, 2021). Through cooperation/collaboration, the cross-fertilization and recombination of ideas can be accelerated among stakeholders. Perhaps the most important benefit of cooperation/collaboration is that it allows for the sharing of relational-specific resources (Dyer & Singh, 1998). In summing up the literature, we opine that cooperation/collaboration will present opportunities for seaport–hinterland networks stakeholders (Nguyen et al., 2021).

Cooperation and collaboration also extend beyond individual stakeholder groups to the seaport–hinterland network itself (Arendt, 2020; Chen et al., 2016). Such cooperation/collaboration is driven by the nature of interconnectedness between the various components/nodes of the seaport–hinterland network (e.g., other ports or alternate port routes/modal options) (Jeevan et al., 2021). Here, cooperation and collaboration offer the seaport–hinterland network various benefits such as (i) efficiency improvements and (ii) service improvements (Ang, 2008). All may have costs/financial implications.

Drawing from the literature which espouses that competition not only emanates from rivalries but, also, from the nature of inter-organizational relationships within networks (Gimeno, 2004; Gnyawali & Madhavan, 2001), it is important to recognize that cooperation/collaboration can be significantly jeopardized by competition and opportunistic behavior among the various actors/stakeholders or among various nodes within the seaport–hinterland network itself (Heaver, Meersman, & Van de Voorde, 2001; Song, 2003). In fact, cooperation/collaboration can be construed as an antecedent to competition (see Choi, Kim, & Kim, 2022). Seaport–hinterland actors and the various components of the seaport–hinterland network (e.g., modal options) may be induced to explore various cooperative/collaborative mechanisms to either counter compete or to compensate for resources expended due to competition (see Eisenhardt & Schoonhoven, 1996). This phenomenon is referred to as '*co-opetition*' (Asadabadi & Miller-Hooks, 2018, 2020; Sdoukopoulos & Boile, 2020; Song, 2003; Tsai, 2002).

Considering the existence of overlapping relationships, competition within *co-opetitive* seaport-hinterland networks is likely to be more intense because of increased opportunities to access and observe the action of network members (Hsieh & Vermeulen, 2014; Tsai, Kuo-Hsien, & Chen, 2011). However, insights gleaned from within networks tend to be more advantageous because, in working closely and sharing resources, 'partners' can more readily and impeccably gain insight into and understanding of the foundational intentions and mechanisms driving the behavior of their *co-opetitive* partners (Roketskiy, 2018). In fact, because of proximity, *co-opetitive* partners are more able to glean insights which are more explicit, more in-depth, and, therefore, less misleading than publicly gleaned knowledge which is usually only available to non-networked firms (Choi et al., 2022). On this basis, we opine that there is sufficient justification for actors/stakeholders and the various components of the seaport–hinterland network (e.g., modal options) to expend efforts developing *co-opetitive* competencies.

On this basis (and despite potential competition), there is strong interest in enhancing and sustaining cooperation/collaboration among actors/stakeholders and various components of the seaport–hinterland network, on the basis of which the potential to enhance the quality of joint operational and strategic decision making is of relevance (Der

Horst, Martijn, & De Langen, 2008; Notteboom, 2008). From a client perspective (shippers/consignee), such joint decisions are expected to lead to cost advantages. However, these joint operational and strategic decisions (which we refer to as '*joint seaport–hinterland network decisions*') will depend on various transport characteristics and related factors. These include cargo value, transportation mode (e.g., truck, rail, barge), cargo accommodation type for transport (consolidated or deconsolidated), and distance carried. They will also depend on other factors such as dwell-time of the stored container, the type and amount of additional services required, and so on. Thus, with these transport characteristics in mind, we set out in this study to develop a flexible cost model to simulate a set of Transportation Strategies (TS) which, we opine, is best placed to aid clients to jointly choose their services providers (seaport/dry port/carrier) across the seaport–hinterland network. Thus, we state our Research Question (RQ) as follows:

RQ: How do transport characteristics drive joint seaport–hinterland network decisions in terms of total container shipping cost?

To address this research question, the rest of the paper is structured as follows. In section 2, a review of previous container shipping models is undertaken. In section 3, we briefly articulate the Brazilian context of our study; we address our methodology, and the formulation of the cost model. Results are presented in section 4 and discussed in section 5. The paper concludes in section 6 by elaborating on the study's theoretical and practical implications.

2. Literature review

There is previous research that has contributed to our understanding of seaport–hinterland networks. We briefly review this literature to develop a concise and cohesive appreciation of the state of existing knowledge on modeling in seaport–hinterland networks (see Paul & Criado, 2020; Torraco, 2016).

Some studies showing the state of the knowledge in this area include those of Rodrigue and Notteboom (2012), Bask, Roso, Andersson, and Hämäläinen (2014), El Hassan (2016), Wang, Monios, and Zhang (2020), and Jeevan et al. (2022). Among these studies, some have focused on model building and application issues which we briefly review. An example of such a model was developed by El Hassan (2016). Taking into consideration the need to optimize the design and configuration of hinterland logistics network infrastructures and services that takes into consideration potential cost differences, a model is proposed that results in service class-based shipping frequencies. Tsao and Linh (2018) employed a nonlinear optimization approach to present a continuous approximation model to facilitate optimized benefits related to carbon emissions in seaport–dry port networks. The outcome of their study suggests the potential reduction of carbon costs from road transportation.

Shan, Peng, Liu, Yao, and Bin (2020) presented a mathematical model for optimizing both locations of inland depots and schedules of container routes. The focus of this model was to determine a formulation of the optimal conditions for pricing the repositioning of empty containers. On the other hand, Kurtuluş (2022) proposed a network design model to optimize total costs and environmental impact (specifically, emissions) with dry port location and allocation decisions, including empty container relocation and volume discount. Findings from the study suggested the possibility of total emissions being reduced to amounts lower that those emerging from minimized costs. The model demonstrated that a change of transportation mode only served as an initial step towards the reduction of emissions prior to any adoption of rail transportation technology. Furthermore, the study pointed out that the reduction of costs of rail transportation served as the most effective means of reducing carbon emissions.

Despite all such highly pertinent contributions of these proposed models, it is our observation that some of these studies have not considered joint operational and strategic decisions. Certainly, as shown in Shan et al. (2020) and Kurtuluş (2022), the clients' perspective appears not to have been incorporated into the modeling.

There are also other studies that have contributed to our understanding of seaport–hinterland networks that are focused on modeling seaport–hinterland cost optimization. These studies include Janic (2007), Iannone and Thore (2010), Iannone (2012), Crainic, Dell'Olmo, Ricciardi, and Sgalambro (2015), Kapetanis, Psaraftis, and Spyrou (2016), Tran, Haasis, and Buer (2017), and Qiu and Lam (2018).

For example, in Janic (2007), a model for internal and external costings of intermodal and road freight transport networks was proposed. The focus of this model was to facilitate the calculation of combined internal and external costs of transport networks taking into consideration European Union (EU) social policies. In Iannone and Thore (2010), the focus was on advancing a model able to facilitate the minimization of total logistics costs over the entire seaport–hinterland distribution network. The primary aim of their model was to assess potential benefits in specific container routing. Iannone and Thore's (2010) study was later extended by Iannone (2012). The focus of this study was to set out an optimizing model focused on economic analysis. One key attribute of the Iannone (2012) model was the minimization of logistic costs. Findings showed the potential for enhancing seaport–hinterland networks' competitiveness through the combined effect of extended gateways, customs facilities, and improved railway connectivity.

Further seaport–hinterland cost optimization studies have since followed. Crainic et al. (2015), for example, developed a mixed integer programming model able to minimize overall logistic cost based on optimal route definitions and scheduling. Kapetanis et al.'s (2016) study advanced an analytical parameterized tool able to support constant assessment among other parameters of the costs associated with alternative modes of transportation. In Tran et al. (2017), the focus was on optimizing container capacity to minimize costs. Findings from this study showed that ship costs and port costs constituted less than a third of total costs. Findings also suggested that employing more ports while increasing the distance of travel (and resulting in higher shipping costs) were likely to result in lower costs associated with hinterland/port distribution. In Qiu and Lam (2018), studying how shared transportation systems can bring significant profit improvements to dry ports as well as cost savings for shippers/consignees, they employed a bi-level model able to show how major profit enhancements (and cost savings) could be derived.

Other optimization models have also been proposed, which mainly focus on the environmental impacts of seaport–hinterland networks (see Kurtuluş, 2022; Lättilä, Henttu, & Hilmola, 2013; Oey & Setiawan, 2017), and for storage of containers traveling both inbound (Kim & Kim, 2007) and outbound (Qiu, Lam, & Huang, 2015).

While the above reviewed literatures does much to summarize the current progress in developing cost-related models for seaport–hinterland decisions in container shipping, the diversity of the models advanced in the literature help highlight existing heterogeneity and, arguably, the need for standard models for seaport–hinterland decisions in container shipping that may be more widely applied and assessed for viability. For example, we are not aware of models focused on the joint choice of service providers by clients to aid seaport–hinterland decisions in container shipping. We are also not aware of modeling which specifically focuses on the Brazilian context. This is despite the country being among the top 20 container-handling economies in the world and, by implication, a major global player in maritime cargo transportation (UNCTAD, 2019). These realities serve as justification for our present study set within the context of Brazil, which aims to propose a cost model able to facilitate joint multi-stakeholder seaport–hinterland network decisions between shippers and consignees (clients), focused on choosing a transportation strategy that minimizes the total costs incurred in container shipping.

3. Methodology

3.1. The Brazilian context

The present study focuses on the joint choice of service providers by clients in the seaport–hinterland network within São Paulo state, Brazil.

Brazil is the ninth largest economy in the world, with a GDP of US $7.4 trillion. In 2020, Brazil's export revenue was approximately US $209 billion, while its import value was approximately US$158 billion (MDIC, 2021; World Bank, 2020). Brazil is also the twentieth largest container-handling economy in the world, handling more than 10 million TEUs (20-ft-equivalent units of cargo capacity) per year, with the Santos seaport hub in São Paulo state being responsible for 45% of this volume (ANTAQ, 2022; UNCTAD, 2019). Within Brazil's hinterland, there are currently 56 dry ports (Economy Ministry, 2020). Seaport–hinterland connectivity in Brazil is dominated by the road network (65% of the transport share). Accordingly, it is recognized in the literature that Brazil has a low rail network density (ANTF, 2019).

The State of São Paulo is Brazil's industrial and economic center. São Paulo has approximately 46 million people living across 645 cities, and a GDP of US$ 603 billion (IBGE, 2021). This makes it the twenty-first largest economy in the world and the third largest economy in South America. In 2019, exports from São Paulo amounted to approximately US$51 billion while imports amounted to approximately US$59 billion. São Paulo has the largest airport (Guarulhos) and seaport (Santos seaport) in Latin America (São Paulo, 2021), with a high concentration of industries and logistics activities throughout the seaport–hinterland. However, despite the high volume of containerized cargo transported, the 24 dry ports spread across 16 cities in São Paulo are not directly connected with Santos seaport. In this case, to deliver or receive a container by railway, it is required to use one of the seven multimodal facilities to then transship the container from the train to a truck. Additionally, São Paulo's seaport–hinterland faces other issues such as the congestion in seaport neighborhoods, intense traffic on the roadways, bureaucratic customs process, and security of the cargo (Rodrigues Thiago de, Mota, Ojiako, & Dweiri, 2021). Fig. 1 represents the seaport–hinterland of São Paulo.

3.2. The study

As in the case of similar prior studies such as Iannone (2012), Lättilä et al. (2013), and Tran et al. (2017), we compare three scenarios, recognizing a total of seven container transportation strategies (TS) as optimizable for cost reduction. Evaluation is undertaken using discrete event simulation. Simulation is chosen as the preferred research approach as real life experimentation was not possible (Kasanen, Lukka, & Siitonen, 1993). Banks, Carson, Nelson, and Nicol (2005) also opine that simulation is particularly suited to developing new designs (or, in our case, new models). Arguably, both analytical modeling and simulation are suitable for our purpose. However, we deemed simulation preferable, as it allows for greater flexibility in terms of incorporation of multiple agents that may impact the decision making (Lättilä et al., 2013).

Three questions are considered jointly in the model: (i) the decision to perform all container logistics services in seaport or dry port; (ii) the decision to transport the container cargo by road or by multimodal transportation (truck and rail); and (iii) to deliver/receive the container consolidated or deconsolidated.

For the simulations, seven transportation strategies (TS) were tested, based on Tran et al. (2017), Fazi and Roodbergen (2018), and

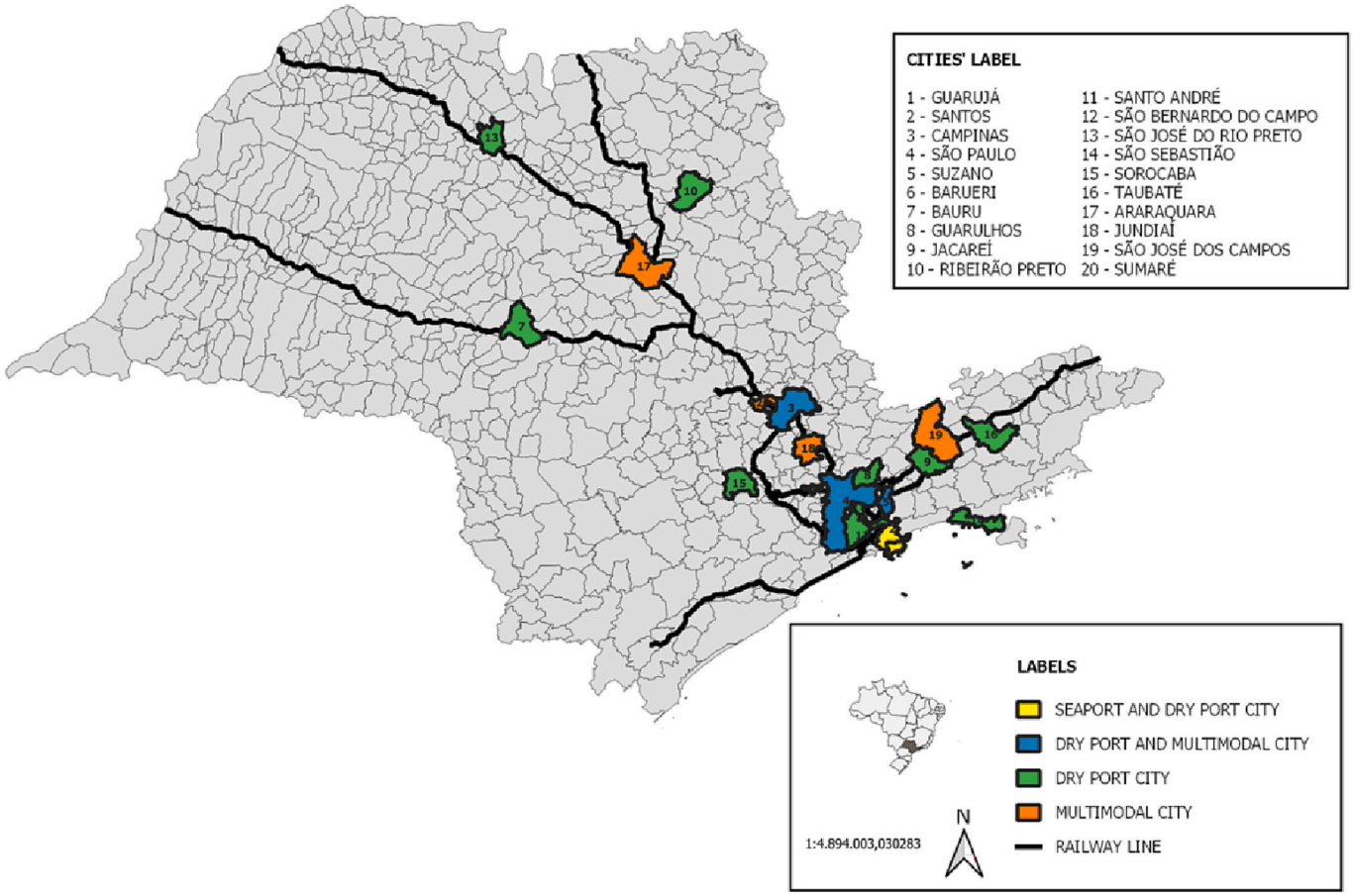


Fig. 1. Seaport–hinterland network of São Paulo.

Sdoukopoulos and Boile (2020):

TS1: when the container is delivered consolidated (con) to the client (k) by truck (t) from the seaport (i), and then returns empty (empt) to the same seaport $\left(i \xrightarrow{t} k_{con}; k \xrightarrow{t} i_{empt}\right)$.

TS2: when the container is delivered consolidated to the client by multimodal transport, rail (r) and truck (t), from the seaport, and then returns empty to the same seaport $\left(i \xrightarrow{r,t} k_{con}; k \xrightarrow{t,r} i_{empt}\right)$.

TS3: when the container is deconsolidated (deco) in the seaport and delivered by truck to the client from the seaport $\left(i \xrightarrow{t} k_{deco}\right)$.

TS4: when the container is delivered consolidated from a seaport to a dry port (j) by truck, then delivered to the client consolidated by truck, and then returns empty to the same dry port $\left(i \xrightarrow{t} j_{con}; j \xrightarrow{t} k_{con}; k \xrightarrow{t} j_{empt}\right)$.

TS5: when the container is delivered consolidated from a seaport to a dry port by multimodal transport (rail and truck), then delivered consolidated to the client by truck, and then returns empty to the same dry port $\left(i \xrightarrow{r,t} j_{con}; j \xrightarrow{t} k_{con}; k \xrightarrow{t} j_{empt}\right)$.

TS6: when the container is delivered consolidated from a seaport to a dry port by truck, to then be deconsolidated in the dry port and delivered to the client by truck $\left(i \xrightarrow{t} j_{con}; j \xrightarrow{t} k_{deco}\right)$.

TS7: when the container is delivered from a seaport to a dry port by multimodal transport (rail and truck), to then be deconsolidated in the dry port and delivered to the client by truck $\left(i \xrightarrow{r,t} j_{con}; j \xrightarrow{t} k_{deco}\right)$.

In Fig. 2, we show the schema of the simulation scenarios.

The cost model is applicable to both import and export container shipping processes; however, as dry ports in Brazil focus mainly on cargo import, the case study is limited to the import process (Rodrigues Thiago de et al., 2021). The cost database was obtained from multiple sources, having as reference the tariff for a 40-ft dry container, as follows. For road cost, we obtained data from the National Association of Cargo Transport and Logistics (NTC, 2021), from resolution 5.867 of the National Agency of Ground Transport (ANTT, 2021), and from the study of Araújo Maria da Penha, de Mello Bandeira, and Campos (2014). The rail fees were based on the average of tariff charged in 2020 for a 40-ft dry container by the two rail carriers operating transport containers in São Paulo state. Dry port fees were based on the service average charges of seven companies responsible for the management of 11 dry ports in São Paulo. In the same way, we considered the average of container service tariff for the year 2021 of the three most important seaports located in the Santos hub and six-line operators/ship-owners. The road distances considered in the model were assessed by Google Maps. The simulations and data analysis were run in Python 3.8.5 and Microsoft Excel (2010), and the sensibility analysis of the TS was built in QGIS 3.16.

3.3. Cost model

The cost model considers a set of seaports (i), dry ports (j), and customers (k) for the simulation, testing all facilities as options for the import process. In order to answer the three main questions stated above, the cost model was defined as options for the customer to decide: i) to import direct from the seaport (i) or to use a dry port (j); ii) to choose the Transportation Mode (TM) as truck (t) or rail (r); and iii) to define the Cargo Accommodation Type for transport (CT) referring to the way that the cargo is carried, as consolidated (con), deconsolidated (deco), or empty (empt).

Based on the seven transportation strategies, 10 cost variables related to container shipping in the seaport–hinterland network was taken into consideration (Rodrigues, Mota, Pinto, & Araújo, 2020; Tran et al., 2017). These cost variables are as follows: (i) Terminal Handling Cost (HC); (ii) Storage Cost (SC); (iii) Value-Adding Service Cost (VA); (iv) Customs clearance cost (CC); (v) Inventory cost (IC); (vi) Demurrage and detention costs (DDC); (vii) Container leasing cost (CLC); (viii) Delivery cost (DEC); (ix) Back empty container cost (BECC); and (x) Emission cost (EMC). Aggregating these variables, the Total Logistic Cost (TLT) is measured, varying according to the Transportation Strategy simulated. The indices and the main variables of the cost model follow in Table 1. Other variables and the parameters are described in Appendix 1.

3.3.1. Terminal handling cost (HC)

Terminal handling cost is based on the number of boxes moved at the seaport (i) or dry port (j) ($box_{i,j}$), the terminal handling charges ($thc_{i,j}$), and the number of handling moves undertaken in each transportation strategy ($qhm_{TS_{i,j}}$) (Crainic et al., 2015; Tran et al., 2017). After being discharged from the vessel, the container is positioned in the seaport's yard, and any subsequent handling of it will incur charges. If the customer chooses to receive the container directly from the seaport, one handling is required to load it onto the truck. However, if the customer chooses to import through a dry port, at least three handlings are required: one to load the container onto the truck or rail, another to discharge the container in the dry port's yard, and a final one to load the container onto the truck for delivery to the customer.

$$HC_{TS} = \sum_{\substack{i \in \mathrm{SEA} \\ j \in \mathrm{DRY} \\ TS \in \mathrm{STS}}} thc_{i,j} \times box_{i,j} \times qhm_{TS_{i,j}} \tag{1}$$

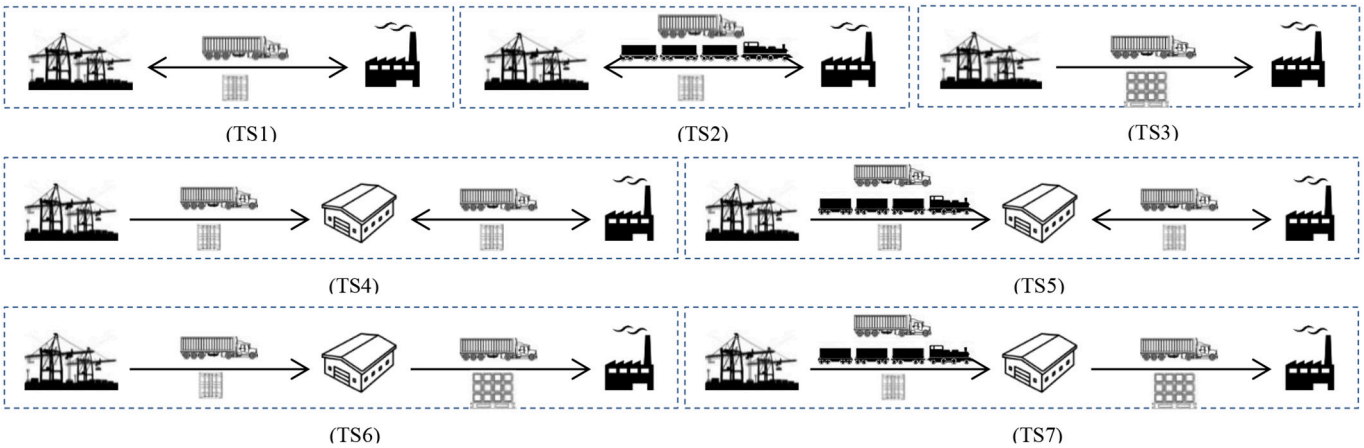


Fig. 2. Transportation Strategy simulations.

Table 1
Indices and variables of the cost model.

Indexes and variables	Description	Assumptions	References
Indexes			
SEA	Set of seaports (i)		
DRY	Set of dry ports (j)		
CUS	Set of customers (k)		
TM	Set of transportation mode: t – truck; r – rail		
CT	Set of cargo accommodation type: con – consolidated; deco – deconsolidated; empt – empty.		
STS	Set of transportation strategies (TS).	$TS1 = (i \xrightarrow{t} k_{con}; k \xrightarrow{t} i_{empt})$ $TS2 = (i \xrightarrow{r,t} k_{con}; k \xrightarrow{t,r} i_{empt})$ $TS3 = (i \xrightarrow{t} k_{deco})$ $TS4 = (i \xrightarrow{t} j_{con}; j \xrightarrow{t} k_{con}; k \xrightarrow{t} j_{empt})$ $TS5 = (i \xrightarrow{r,t} j_{con}; j \xrightarrow{t} k_{con}; k \xrightarrow{t} j_{empt})$ $TS6 = (i \xrightarrow{t} j_{con}; j \xrightarrow{t} k_{deco})$ $TS7 = (i \xrightarrow{r,t} j_{con}; j \xrightarrow{t} k_{deco})$	Tran et al. (2017); Fazi and Roodbergen (2018); Sdoukopoulos and Boile (2020).
SAS	Set of additional services (kas_q) in seaports (i) and dry ports (j).	kas_1 = positioning kas_2 = deconsolidating kas_3 = consolidating kas_4 = weighting	Defined by the authors
Variables			
HC_{TS}	Terminal handling cost in seaport (i) and dry port (j) by Transportation Strategy (TS).		Janic (2007); Iannone (2012); Crainic et al. (2015); Tran et al. (2017); Tsao and Linh (2018).
SC_{TS}	Storage cost in seaport (i) and dry port (j) by Transportation Strategy (TS).		Iannone (2012); Qiu et al. (2015); Crainic et al. (2015); Oey and Setiawan (2017); Qiu and Lam (2018); Tsao and Linh (2018).
VA_{TS}	Value-adding service cost in seaport (i) and dry port (j) by Transportation Strategy (TS).		Janic (2007); Monios (2011); Bask et al. (2014); Crainic et al. (2015); Jeevan, Chen, and Cahoon (2017).
CC_{TS}	Customs clearance cost in seaport (i) and dry port (j) by Transportation Strategy (TS).		Iannone and Thore (2010); Crainic et al. (2015).
TLT_{TS}	Total logistic time by Transportation Strategy (TS).		Fazi and Roodbergen (2018); Iannone (2012); Talley and Ng (2017); Tran et al. (2017).
IC	Inventory cost.		Janic (2007); Iannone (2012); Kapetanis et al. (2016); Oey and Setiawan (2017); Tsao and Linh (2018).
DDC_{TS}	Demurrage and detention cost by Transportation Strategy (TS).		Tongzon (2009); Yeo et al. (2014); Crainic et al. (2015); Oey and Setiawan (2017); Rezaei et al. (2019); Hsu, Lian, and Huang (2020).
CLC_{TS}	Container leasing cost by Transportation Strategy (TS).		Tran et al. (2017)
DEC_{TS}	Delivery cost by Transportation Strategy.		Janic (2007); Tongzon (2009); Iannone and Thore (2010); Iannone (2012); Crainic et al. (2015); Larranaga, Arellana, and Senna (2016); Tran et al. (2017); Tsao and Linh (2018); Jiang et al. (2020).
$BECC_{TS}$	Back empty container cost by Transportation Strategy (TS).		Iannone and Thore (2010); Iannone (2012); Song and Dong (2012).
EMC_{TS}	Emission cost by Transportation Strategy (TS).		Lättilä et al. (2013); Tran et al. (2017); Tsao and Linh (2018).
TLC_{TS}	Total logistic cost by Transportation Strategy (TS).	Aggregation of all costs.	Defined by the authors.

3.3.2. Storage cost (SC)

The storage cost is proportional to the quantity of containers stored at the seaport (i) or dry port (j) ($box_{i,j}$); the storage charge per cargo consolidated ($scc_{i,j}$) or deconsolidated ($scd_{i,j}$); and the dwell-time of the cargo in the seaport or dry port facility ($doc_{i,j}$) (Qiu et al., 2015; Tsao & Linh, 2018). The model simulates the storage cost depending on the cargo accommodation type (CT), varying from consolidated cargo (*con*) in the yard (when $TS = 1, 2, 4$ *or* 5) to deconsolidated cargo (*deco*) in the warehouse (when $TS = 3, 6,$ *or* 7). Some assumptions of the model are that storage free time in the seaport or dry port's yard or warehouse is not considered. Additionally, a minimum rate is considered applicable for consolidated ($msrc_{i,j}$) and deconsolidated ($msrd_{i,j}$) cargo when the storage charge per cargo ($scc_{i,j}$; $scd_{i,j}$) is less than the minimum charged ($msrc_{i,j}$; $msrd_{i,j}$).

$$SC_{TS} = \sum_{\substack{i \in SEA \\ j \in DRY \\ TS \in STS}} scc_{i,j} \times scd_{i,j} \times doc_{i,j} \times box_{i,j} \quad (2)$$

The scc_{TS} and scd_{TS} can be measured as a percentage of the cargo value (*cv*) for consolidated ($src_{i,j}$) and deconsolidated ($srd_{i,j}$) cargo. In this model, the cargo value (*cv*) is measured through a summation of the CIF (Cost, Insurance, and Freight) rate (*cifr*), the cargo weight (*cw*), and the dollar rate (*dor*) (Lättilä et al., 2013; Zhang & Schramm, 2020). Additionally, if the cargo is deconsolidated, it must be stored and delivered as a whole, not partially.

The storage cost is dependent on the dwell-time of cargo at the seaport or dry port facility ($doc_{TS_{i,j}}$). To estimate the dwell-time, the variable *ttd* is used, which represents the time between container discharge at the seaport and its arrival at the customer facility. It will

vary from 2 days until 60 days in the simulations. The $doc_{TS_{i,\ j}}$ is estimated by deducting the transit time of the cargo and the documentation time (custom process) required for releasing the cargo ($dtd_{i,\ j}$) from *ttd*. The transit time may vary based on the transit time from the seaport to the client ($ttsc_{TS_{t,\ r}}$), from the seaport to the dry port ($ttsd_{TS_{t,\ r}}$), and from the dry port to the client ($ttdc_{TS_t}$), according to the Transportation Strategy, as outlined in eq. (3).

$$doc_{TS_{i,j}} = ttd - \sum_{\substack{t,r\in \mathrm{TM} \\ TS\in \mathrm{STS}}} ttsd_{TS_{t,r}} - ttsc_{TS_{t,r}} - ttdc_{TS_t} - dtd_{i,j} \quad (3)$$

3.3.3. Value-adding service cost (VA)

In addition to offering services such as loading/unloading, storage, customs, and warehousing, dry ports and seaports are expected to provide a range of value-adding services—for example, sorting, mixing, blending, making, barcoding, packing, and labeling (Jeevan et al., 2017). Such activities are likely to be less cost-intensive in dry ports (Bask et al., 2014). As the range of value-adding services available vary among the seaports and dry ports, the model will consider a set of value-adding services ($SAS = kas_1, kas_2, \ldots, kas_N$) and the respective charges ($\mathrm{cas}_{kas_{i,\ j}}$). For instance, if the decision maker decides to deconsolidate the cargo in the seaport, value-adding services as cargo weighting should be considered in the model. In this way, value-adding service cost is assessed as follows:

$$VA_{TS} = \sum_{\substack{i\in \mathrm{SEA} \\ j\in \mathrm{DRY} \\ kas\in \mathrm{SAS}}} \mathrm{cas}_{kas_{i,j}} \times box_{i,j} \quad (4)$$

3.3.4. Customs clearance cost (CC)

Essentially, there are four different customs control types: (1) automated computerized control, (2) documentary control, (3) physical inspection, and (4) X-ray scanner control (Iannone & Thore, 2010). The customs clearance cost is assessable by considering the positioning for physical inspection cost ($pic_{TS_{i,\ j}}$), x-ray scanner control cost ($xsc_{TS_{i,\ j}}$), and the transshipment customs cost ($tcc_{TS_{i,\ j}}$) for moving transfer containers from seaports to dry ports. As customs charges for automated computerized control and documentary control are the same and invariable for all custom offices, this was not considered in the model.

$$CC_{TS} = \sum_{\substack{\mathrm{TS}\in \mathrm{STS} \\ i\in \mathrm{SEA} \\ j\in \mathrm{DRY}}} \left(pic_{TS_{i,j}} + xsc_{TS_{i,j}} + tcc_{TS_{i,j}} \right) \times box_{i,j} \quad (5)$$

3.3.5. Total logistic time (TLT)

Time is one of the most important factors in logistics decision making. Clients usually require their cargo to be delivered as quickly as possible. To assess the '*time cost*', it is necessary to measure the Total Logistics Time (TLT). This is the time expended for the cargo at each step of the import/export process, also considering the return of the empty container, when applicable. TLT is defined as the dwell-time of the cargo at a seaport or dry port facility ($doc_{i,\ j}$), which includes the documentation time to release the cargo, $dtd_{i,\ j}$, and the transit times mentioned in eq. (3), including the time to return the empty container ($ttre_{TS_{t,\ r}}$). Hence, the TLT_{TS} is measured by eq. (6).

$$TLT_{TS} = \sum_{\substack{TS\in \mathrm{STS} \\ t,r\in \mathrm{TM}}} ttd + ttre_{TS_{t,r}} \quad (6)$$

3.3.6. Inventory cost (IC)

Inventory cost may be expressed by aggregating (i) opportunity (*or*) and depreciation (*der*) costs, which is the inventory in transit holding costs depending on the cargo value (*cv*); (ii) the interest rate (opportunity and depreciation rates per day); and (iii) the time between the container discharges at the seaport and its arrival at the customer facility (*ttd*) (Iannone & Thore, 2010; Talley & Ng, 2017).

$$IC = ttd \times cv \times (or + der) \quad (7)$$

3.3.7. Demurrage and detention costs (DDC)

Ship-owners typically charge a combined rate that includes demurrage and detention charges, based on the transit time of the container (ttc_{TS}) from when it is discharged at the seaport until it returns empty to the seaport or dry port. This transit time may vary depending on the Transportation Strategy. Moreover, the DDC rate is progressive, with a demurrage and detention fee 1 (*ddf*1) charged after the ttc_{TS} exceeds the demurrage and detention free-time (*ddft*). After ttc_{TS} surpasses double the *ddft*, a demurrage and detention fee 2 (*ddf*2) becomes applicable until the empty container's return. The model makes two main assumptions regarding demurrage and detention: i) if the cargo is deconsolidated in the seaport, no DDC is charged (Fazi & Roodbergen, 2018); ii) The empty container must be returned to the same facility (dry port or seaport) from which it was originally dispatched. In this way, the DDC is measured by eqs. 8 and 9.

If $ttc_{TS} \le ddft$, $DDC_{TS} = 0$

$$\text{If } ddft < ttc_{TS} \le (2 \times ddft), DDC_{TS} = ttc_{TS} \times ddf1 \quad (8)$$

$$\text{If } ttc_{TS} > (2 \times ddft), DDC_{TS} = (ddft \times ddf1) + \{[ttc_{TS} - (2 \times ddft)] \times ddf2\} \quad (9)$$

3.3.8. Container leasing cost (CLC)

Container leasing cost is measured as function of a container leasing charge per day (*lcd*) and the container transit time (ttc_{TS}), since it is discharged from the vessel (full) until returns to the ship-owner (empty) (Iannone, 2012). This cost is measured by eq. 10.

$$CLC_{TS} = ttc_{TS} \times lcd \quad (10)$$

3.3.9. Delivery cost (DEC)

Delivery cost is the charge paid by clients to receive their cargo from a dry port or seaport. This model quantifies the delivery cost based on several factors: the number of boxes delivered ($box_{i,\ j}$), the modal charge ($mc_{t_{con,\ deco}, r_{con}}$), which varies depending on the transportation mode (truck or rail), and the cargo accommodation type (consolidated or deconsolidated); and the delivery distance ($dd_{TS,\ TM}$), which depends on the Transportation Strategy and the Transportation Mode (Lättilä et al., 2013). Carriers also charge a fixed delivery rate ($fdr_{t_{con,\ deco}, r_{con}}$) to cover their services. An additional fee (adc_{TS}) is also charged for the transportation under customs control rate, depending on the Transportation Strategy. Furthermore, an operational trans-shipment rate (otr_{TS}) is applied to transfer the container from the train to the truck if the dry port is not linked to the seaport by rail. The distances vary according to the Transportation Strategy and the location of each facility. Therefore, the delivery cost (DEC_{TS}) is measured in by eq. 11 (Lättilä et al., 2013; Tran et al., 2017).

$$DEC_{TS} = \sum_{\substack{i\in \mathrm{SEA} \\ j\in \mathrm{DRY} \\ k\in \mathrm{CUS} \\ t,r\in \mathrm{TM} \\ con,deco\in \mathrm{CT} \\ TS\in STS}} \left(box_{i,j} \times dd_{TS,TM} \times mc_{t_{con,deco}, r_{con}} \right) + fdr_{t_{con,deco}, r_{con}} + adc_{TS} + otr_{TS} \quad (11)$$

3.3.10. Back empty container cost (BECC)

Shipping companies are usually responsible for providing the required empty container to their customers. When a customer requests to export cargo, a truck transports an empty container from a depot (either a dry port or seaport) to the customer's location to load the cargo. After the consignee receives the container and unloads it, they instruct the shipping company on which depot they should deliver the unit to. This decision, made by the shipping company, is based on the inventory and demand for each depot and container type (Song & Dong, 2012).

With that in mind, the transportation cost to return the empty container is measured by eq. (11). Some assumptions for the simulation are that the empty container should be sent back to the same facility from where it was delivered, through either road or rail transportation modes. In this way, ($dd_{TS,\ TM}$) is the same distance in the route taken for delivering the cargo from the client (k) to the dry port (j) or seaport (i). Additionally, if the cargo is unloaded in the seaport or dry port, no BECC is charged.

$$BECC_{TS} = \sum_{\substack{i\in \mathrm{SEA} \\ j\in \mathrm{DRY} \\ k\in \mathrm{CUS} \\ empt\in CT \\ t,r\in \mathrm{TM} \\ TS\in STS}} \left(box_{i,j} \times dd_{TS,TM} \times \mathrm{mc}_{t_{empt},r_{empt}}\right) + fdr_{t_{empt},r_{empt}} + otr_{TS} \qquad (12)$$

3.3.11. Emission cost (EMC)

A well-designed dry port can bring about environmental benefits by shifting cargo movements from road to rail (Roso et al., 2009). In this model, emission cost is considered as an opportunity cost, given that the government offers financial incentives for reducing CO2 emissions by switching the transportation mode from road to rail. The EMC is a carbon credit for shippers and consignees who make this switch (TS = 2, 5, and 7). The emission cost depends heavily on the delivery distance per transportation strategy (dd_{TS}), the weight of the truck with the cargo (w_t), or the weight of wagon plus cargo (w_r). Additionally, emission costs vary through a marginal emission parameter ($me_{t,\ r}$), which depends on the transportation mode (truck or rail) (Lättilä et al., 2013). As a carbon credit, this opportunity cost is applicable only when the customer uses multimodal transport in part of the TS. With that in mind, the carbon credit is measured by comparing the emissions generated by the transportation strategy that uses multimodal transport ($dd_{TS,\ TM=t,\ r}$) with an option to use the same strategy delivering only by truck ($dd_{TS,\ TM=t}$), thus finding how much carbon was saved through the rail connection. Hence, the total carbon credit saved is converted into financial benefits through a carbon rate (cr) as follows in eq. (13).

$$EMC_{TS} = cr \times \sum_{\substack{TS\in \mathrm{STS} \\ t,r\in \mathrm{TM}}} \left(dd_{TS,TM=t} \times me_t \times w_t\right) - \left(dd_{TS,r} \times \mathrm{me}_r \times w_r\right) - \left(dd_{TS,t} \times me_t \times w_t\right) \qquad (13)$$

3.3.12. Total logistics cost (TLC)

The Total Logistics Cost ($\mathrm{TLC_{TS}}$) function per Transportation Strategy is measured by aggregating the Service Cost (SEC), Time Cost (TIC), and the Transportation Cost (TRC) in eq. (14).

$$TLC_{TS} = \sum_{TS\in \mathrm{STS}} HC_{TS} + SC_{TS} + VA_{TS} + CC_{TS} + IC + DDC_{TS} + CLC_{TS} + DEC_{TS} + BECC_{TS} + EMC_{TS} \qquad (14)$$

3.4. Assumptions for the case study

Some assumptions were required for the simulations in the case of São Paulo. To measure the handling cost, each transportation strategy was analyzed, finding that, for TS1 and TS2, only one handle is required to load the container onto the truck or rail to be delivered. For TS3, as the container is unloaded in the seaport's warehouse, and the positioning is considered as an additional service, no handling cost is charged. For TS4 and TS5, they were considered one handling in the seaport and two handlings in the dry port. Last, TS6 and TS7 were considered one handling in the seaport and one handling in the dry port.

The time taken for some activities along the import process was considered as follows: i) the time to deconsolidate the cargo was not considered, which means that, for TS3, TS6, and TS7, the container is unloaded on the same day as it arrives at the seaport or dry port facility; ii) it was considered to take two days to release the cargo that goes to dry port through Customs Transit Declaration (DTA) as free of storage charge (RFB, 2019); and iii) in the case that the container does a gate-in and gate-out in the same day in the seaport or dry port, no storage is charged.

As the need for additional services varies according to customer requirements, the simulations considered only a set of additional services for each TS as positioning, deconsolidation, consolidation, and weighting. In terms of the customs process, as it is the same in dry ports and seaports, the charges related to the process was not considered in the model. Transshipment customs cost ($tcc_{i,\ j}$) is only charged in TS which use dry ports; the seaport charges to release the DTA and the dry port charges to receive it (Ng, Padilha, & Pallis, 2013).

Some assumptions were also made to measure the time cost (TIC). First, all the road and rail distances were measured by Google Maps, making it possible to estimate the transit times according to average speed ($\mathrm{as_{t,\ r}}$). Second, the transit time considered the eight working hours allowed by the Brazilian Legislation. Thus, if a transit time to deliver a container exceeds eight hours, another day of transit was considered. Third, the operational time to load and discharge on the truck or rail, as well the time to consolidate and deconsolidate the cargo, is not considered in the model. Fourth, it is assumed that the customer will return the empty container to either the dry port or the seaport on the same day that they receive it. Last, the congestion in the seaport and the dry port, which may cause delays to remove the container, was not considered in the model.

The simulations assumed the usage of 40-ft container boxes and a road carrier profit margin of 10% (Araújo Maria da Penha et al., 2014). The transportation cost was measured considering the dry ports, seaports, and multimodal facilities available in São Paulo state. Thus, the transportation cost from the Santos hub seaport to all customers (here represented by the cities) was calculated, as well as the cost from the seaport to all dry ports and multimodal facilities. Because, in São Paulo, there is no dry port directly connected with the seaport, we decided to use multimodal facilities as an option in part of the transport for consolidated cargo. In this scenario, the container is transported to a multimodal facility, where it is unloaded from a train and loaded onto a truck for delivery to either a dry port or to the customer. This operation is charged by the operational transshipment rate ($\mathrm{otr_{TS}}$).

For deliveries within the same city or neighboring cities, where the distance to the dry port or seaport is less than 10 km, the fixed delivery rate ($fdr_{t_{con,\ deco,\ empt},\ r_{con,\ empt}}$) plus 10 km of distance for modal charge ($mc_{t_{con,\ deco,\ empt},\ r_{con,\ empt}}$) were considered. Finally, as there is no carbon credit policy in Brazil to encourage shippers and consignees to use railways instead of roadway connections, we have simulated a scenario considering a carbon credit policy to understand if it is an option for policy makers in Brazil. For the simulations, we have considered a carbon price of R $280.00 (US$50.00) per ton of CO2 saved (EDF, 2020).

4. Results

The simulations sought to address the research question by identifying how transport characteristics drive jointly made seaport–hinterland network decisions in terms of cost, taking into consideration the customer's and service provider's locations, the import time since the cargo is discharged in the seaport's yard, the kind of transport (truck or rail), the mode of cargo transportation (consolidated or deconsolidated), and the cargo value. The first simulation (SIM1) ran a cost model with a cargo value of R$415,580.00 (approximately US$74,000.00) (Lättilä et al., 2013; Zhang & Schramm, 2020). This varied the total import time until client receipt of the cargo (TTD) from 2 days up to 60 days, with a view to identifying the impact of those characteristics across the seven TS defined within the simulation as options for consignees to import their cargo. These TS were earlier set out at the beginning of the methodology section as TS = 1 to 7. Fig. 3 presents the result of the seven discrete event simulations, highlighting the most cost-efficient transportation strategy for each city in São Paulo state.

In cases where consignees need their imported cargo as quickly as possible (as shown in the simulation by TTD = 2), the cost-efficient TS

are TS1, TS2, and TS3, where the cargo is directly delivered to the client from the seaport (Fig. 3(i)). This result demonstrates that, in cities near the Santos hub seaport, delivery of containers in a consolidated form, followed by return of empty containers, is the cost-efficient option (TS1). However, as the distance from the seaport increases, deconsolidation of cargo in seaport becomes the cost-efficient option (TS3). Last, multimodal transportation appeared cost-efficient for some cities, especially the ones near the Araraquara's multimodal facility (city number 17 in Fig. 1 and 356 km from the Santos seaport). When TTD = 5, particularly due to customs delays, TS1 becomes the cost-efficient option for most cities in São Paulo (see Fig. 3(ii)). However, TS4 emerges as an option for cities near the dry port of São José do Rio Preto (city number 13 in Fig. 1 and 518 km from the Santos seaport). TS5 emerges as an option for cities near the dry ports of Ribeirão Preto and Bauru (cities numbered 10 and 7 in Fig. 1, far 392 km and 401 km from Santos seaport, respectively). These use the multimodal facility of Sumaré (city number 20 in Fig. 1 and 196 km from the Santos seaport). Last, TS6 starts to become cost-efficient for some cities using dry ports to deconsolidate cargo prior to client delivery.

As the TTD increases from 10 to 15 days, importations of cargo via dry port become most cost-efficient for all São Paulo's cities, specifically through TS4. For TTD = 20, cargo deconsolidation in the dry ports begins to represent a cost-efficient option, especially for cities far from dry port facilities. For TTD = 30, TS4 remains the cost-efficient option, especially for cities near dry port facilities, while TS6 represents the best option for the other cities. Last, as the TTD increases towards 60 days, TS6 becomes the best option for all cities, delivering deconsolidated cargo to the client via dry port.

The second simulation (SIM2) ran the cost model in a manner that varied the total time expended until the client receives the cargo from 2 to 60 days. At the same time, the cargo value was reduced to R $45,000.00 (approximately US$8040.00), following the minimum cargo value considered for storage in Santos' hub port (DP World, 2021; Zhang & Schramm, 2020). The objective here was to understand the impact of

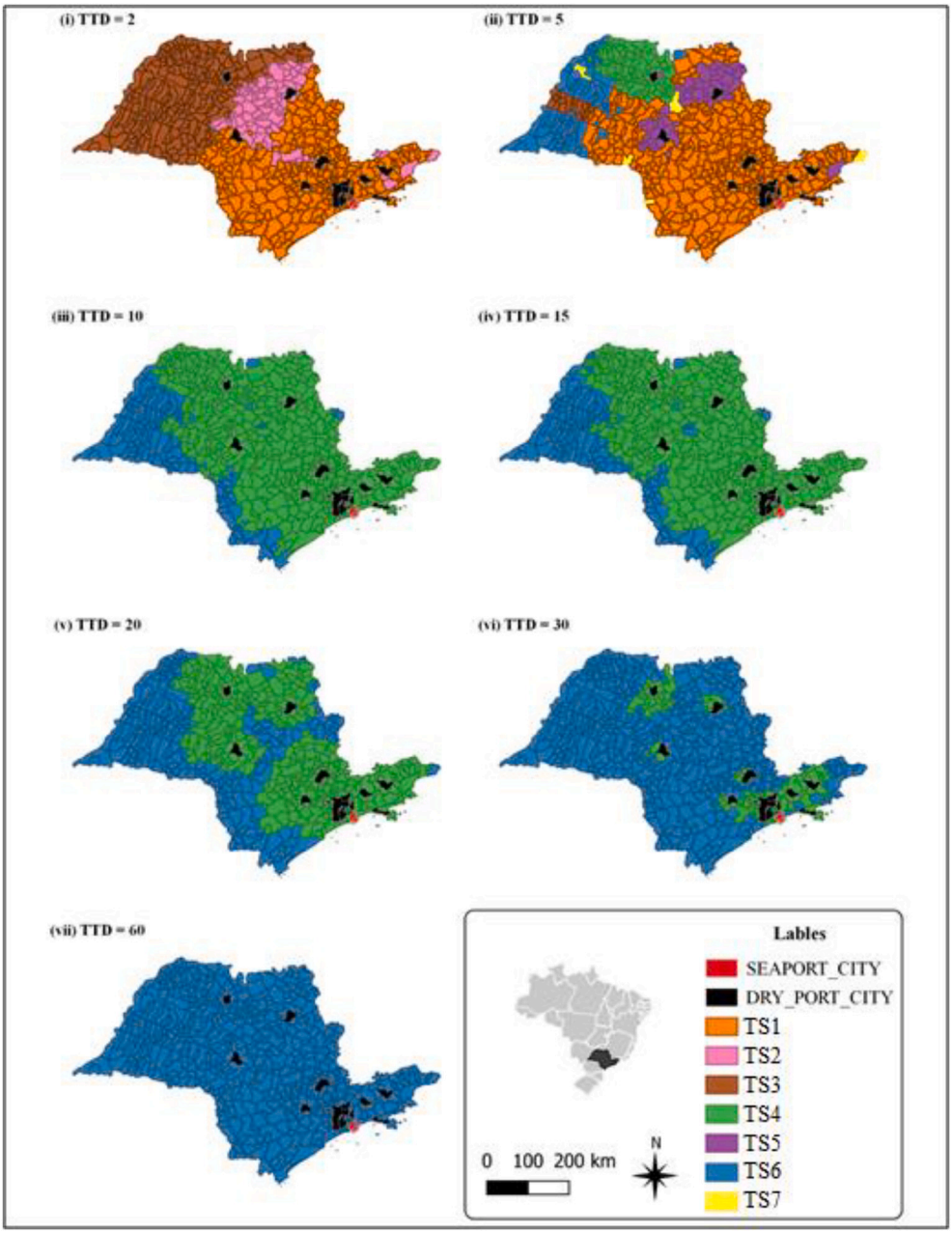


Fig. 3. Results of simulation 1.

the cargo value upon total logistics cost, considering storage costs based on cargo value while also considering minimum storage fees where applicable. The results are shown in Fig. 4.

For TTD = 5, the results were close to those of SIM1, with TS1, TS2, and TS3 as the most cost-efficient options for direct delivery of cargo to the client from the seaport facility. However, for TTD = 5, the scenario changes, with TS1 now appearing as most cost-efficient for most cities. Furthermore, TS3 now appears as the best option for cities far from dry port facilities, with TS4 appearing as best for cities near São José do Rio Preto's dry port (city number 13 in Fig. 1 and 518 km from the Santos seaport). For TTD = 10, the differences between SIM1 and SIM2 were clearer. While in SIM1, cost efficiency for all cities entailed using dry ports, in SIM2, TS1 remains as the best option for most cities, entailing direct seaport-to-client delivery. This finding suggests that, over a longer period of time, seaports remain the cost-efficient option for cargos with low value.

As TTD increased to 15 days, for cargo passing through dry ports, TS6 became the cost-efficient option for all cities. However, for TTD = 20, TS7 emerged as cost-efficient for most cities. This is because, since multimodal transportation takes more time to deliver cargo (considering the transshipment from rail to road), time in storage is reduced. As the cargo has a low value, the minimum storage fee is charged in the dry ports for each 15 days of storage. This additional storage time for TS which use trucks as the transportation mode (as cargo transported using trucks reaches the facility sooner, it requires more storage time) makes for TTD = 20, 35, and 50 days, the TS7 is more cost-efficient than TS6 for a low average difference of R$193.86 (approximately US$35.00). After that, for TTD equal to 30, 60, and so on, TS6 is highlighted as the cost-efficient TS.

The graphs in Fig. 5 compare simulations 1 and 2 in terms of the cost-efficient options per city for each analysis. These options include (i) the kind of logistics operator (dry port or seaport), (ii) the transportation

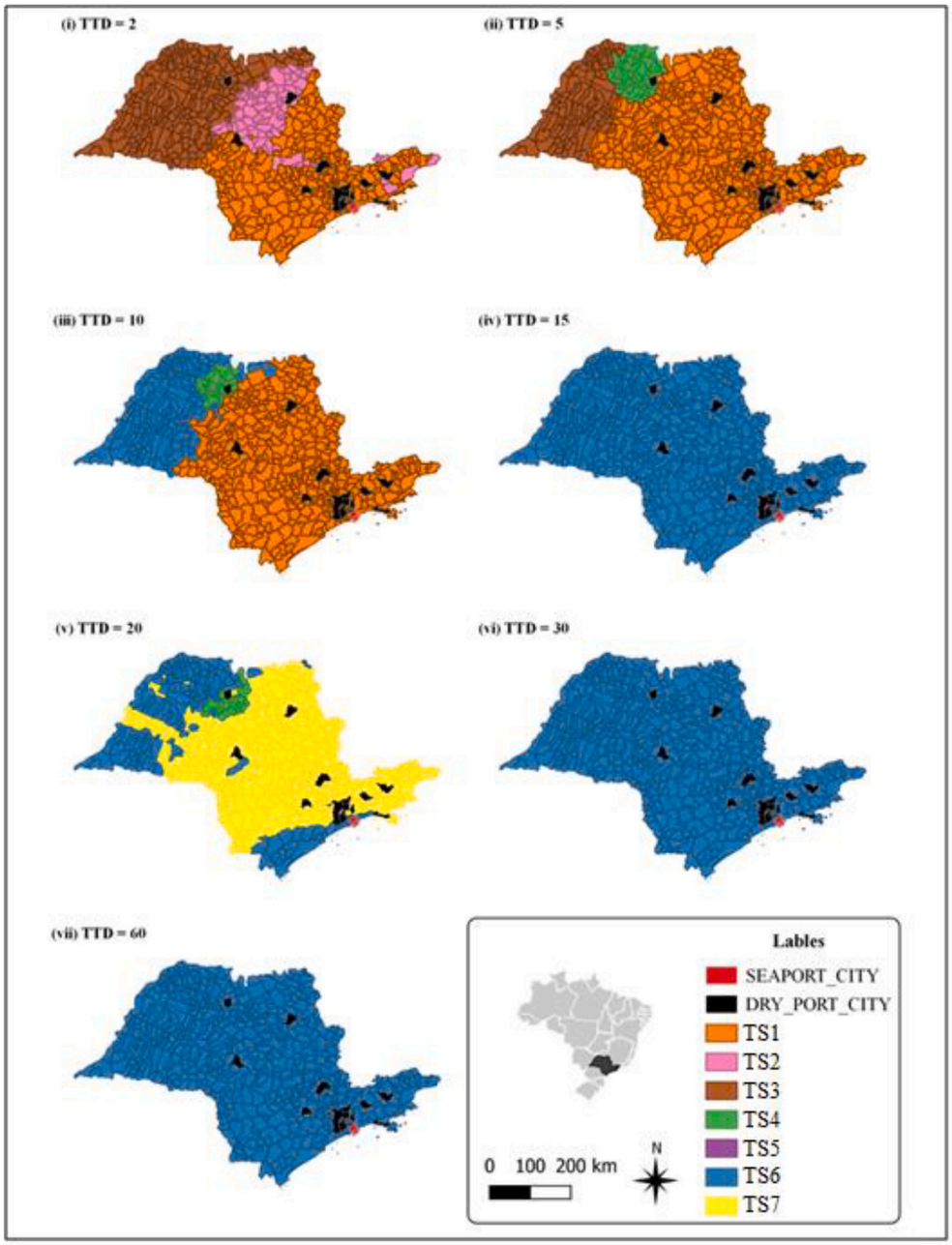
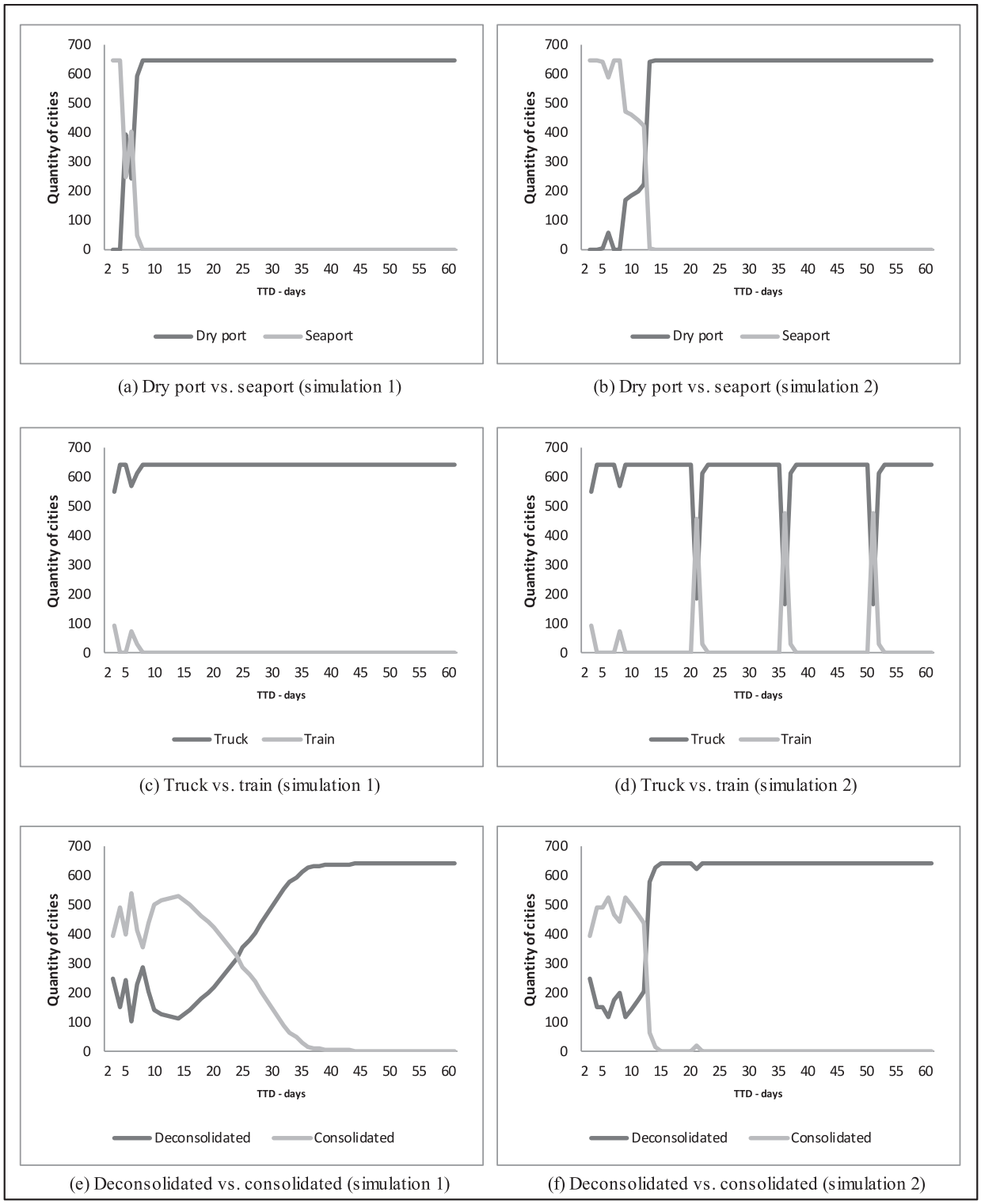


Fig. 4. Results of simulation 2.

mode used (road or rail), and (iii) the cargo accommodation type (consolidated or deconsolidated). The results shown in Fig. 5(a) demonstrate that, for a high cargo value, import through dry port is cost-efficient for most cities in São Paulo when the client requests their cargo after six days. However, when requests are made to reduce that time period, operation through a seaport is recommended in terms of cost. For a low cargo value (Fig. 5(b)), operation through dry ports becomes cheaper for most cities only after 12 days. In this way, as the cargo value reduces and the storage time increases, operation through dry ports becomes cost-efficient. Fig. 5(c) and (d) show that cargo carried by truck was the most cost-efficient option for most cities. This is because there are no dry ports directly connected to the seaport. This makes it necessary to use multimodal facilities to ensure that the cargo reaches a dry port facility (train plus truck). Last, this has the effect of increasing the total logistics cost.

There are, however, some exceptions, for example in the case of low



(a) Dry port vs. seaport (simulation 1)

(b) Dry port vs. seaport (simulation 2)

(c) Truck vs. train (simulation 1)

(d) Truck vs. train (simulation 2)

(e) Deconsolidated vs. consolidated (simulation 1)

(f) Deconsolidated vs. consolidated (simulation 2)

Fig. 5. Comparison between SIM1 and SIM2.

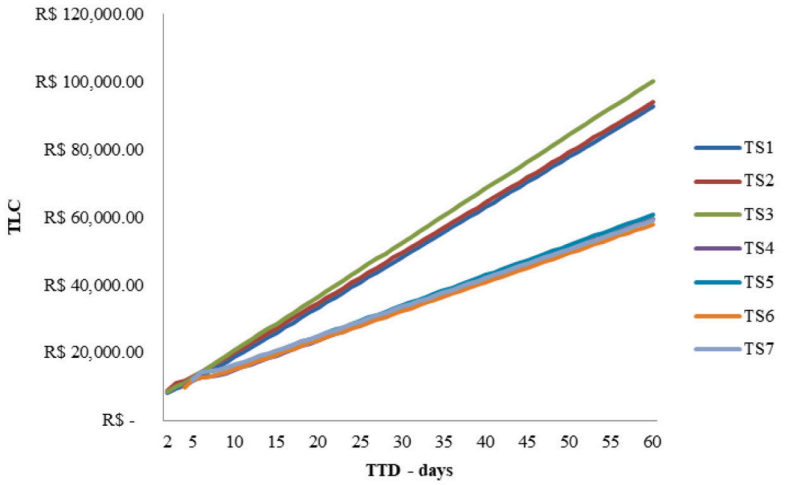
cargo value as shown in Fig. 5 (d) for TTD = 20 and multiples of 15 days as mentioned above. Moreover, for a high cargo value in Fig. 5 (e), it is deemed cost-efficient to transport the cargo consolidated until 24 days for most cities in São Paulo. However, this reduces to 12 days for low cargo values (see Fig. 5(f)). There are two factors which influence these reductions. The first are the demurrage and detention charges and the empty returns, which surpass the deconsolidated storage cost. The second are those costs of additional deconsolidation services over a short time period of time.

Summarizing the results, Fig. 6 presents the variation of the total logistics costs (TLC) by the transit time in each TS for SIM1 (Fig. 6(a)) and SIM2 (Fig. 6(b)). The graphs show that, for high cargo values, dry ports become cost-efficient as the TTD increases. Considering the proposed model, the total logistics cost for all options of TS increases linearly, and at rates of R$1473.00/day (US$263.00/day) for TS1 and TS2; R$1590.00/day (US$284.00/day) for TS3; R$886.00/day (US$158.00/day) for TS4 and TS5; and R$850.00/day (US$152.00/day) for TS6 and TS7.

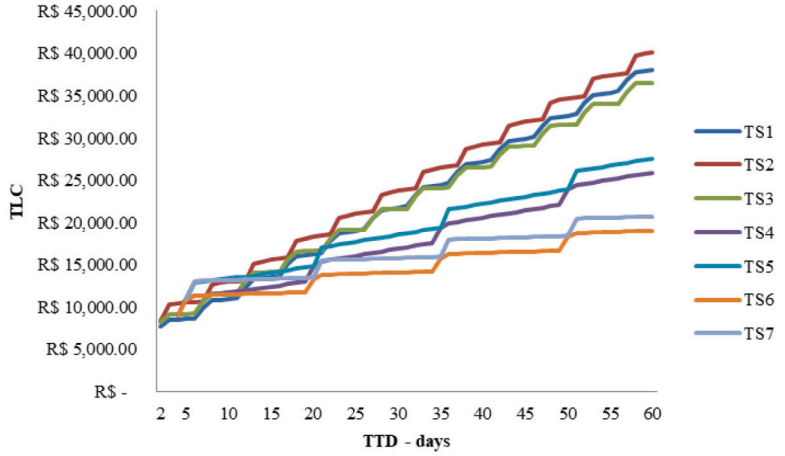
In contrast, for cargos with low values, as in SIM2, the variability of the total logistics cost segregates three groups of TS: (i) TS1, TS2, and TS3 which use seaport to delivery directly, start as the cost-efficient options and become the most expensive options as TTD increases; (ii) TS4 and TS5, which use dry port and deliver the cargo consolidated, returning the empty container; and (iii) TS6 and TS7, which are cost-efficient TS as TTD increases, going through dry ports and delivering the cargo deconsolidated.

Differences of variation between Fig. 6(a) and (b) occur because, in 6 (a), the storage cost for high cargo value surpasses the minimum storage fee, which does not happen in 6(b). This makes the total logistics cost vary according to the minimum storage time, which is 5 days for seaports and 15 days for dry ports. Furthermore, the impact of demurrage and detention fees, and the return of the empty containers, makes the option to deliver the cargo consolidated more expensive. The variability of total logistics cost in SIM2 is approximately R$539.00/day (US $96.00/day) for TS1 and TS2; R$495.00/day (US$88.00/day) for TS3; R $295.00/day (US$52.00/day) for TS4 and TS5; and R$163.00/day (US $29.00/day) for TS6 and TS7.

From another perspective, to simulate the effects of environmental policies to encourage multimodal transportation, two scenarios were tested. Simulation 3 (SIM3) has the same characteristics and inputs as SIM1, adding the emission cost factor (EMC_{TS}); and simulation 4 (SIM4) has the same characteristics and inputs of SIM2, adding the emission cost factor. Table 2 compares the results from the four simulations disaggregating the cost-efficient TS in the three main transport



a) Simulation 1



b) Simulation 2

Fig. 6. Variation of the TLC.

Table 2
Comparison of simulations by the transport characteristic in the cheapest TS.

Transport characteristic	Simulation	Quantity of cities						
		TTD = 2	TTD = 5	TTD = 10	TTD = 15	TTD = 20	TTD = 30	TTD = 60
Dry port	SIM1	–	241	645	645	645	645	645
	SIM2	–	56	200	645	645	645	645
	SIM3	–	248	645	645	645	645	645
	SIM4	–	56	200	645	645	645	645
	SIM1	645	404	–	–	–	–	–
	SIM2	645	589	445	–	–	–	–
	SIM3	645	397	–	–	–	–	–
Seaport	SIM4	645	589	445	–	–	–	–
	SIM1	549	570	645	645	645	645	645
	SIM2	549	645	645	645	187	645	645
	SIM3	488	563	645	645	645	645	645
Truck	SIM4	488	645	645	645	174	645	645
	SIM1	96	75	–	–	–	–	–
	SIM2	96	–	–	–	458	–	–
	SIM3	157	82	–	–	–	–	–
Train	SIM4	157	–	–	–	471	–	–
	SIM1	393	541	516	504	398	119	–
	SIM2	393	525	470	–	22	–	–
	SIM3	393	541	516	504	398	119	–
Consolidated	SIM4	393	525	470	–	22	–	–
	SIM1	252	104	129	141	247	526	645
	SIM2	252	120	175	645	623	645	645
	SIM3	252	104	129	141	247	526	645
Deconsolidated	SIM4	252	120	175	645	623	645	645

characteristics stated. In this way, it shows for how many cities in São Paulo—from a total of 645 cities—it is cheaper to use dry port or seaport as TTD increases in each simulation; for how many cities it is cheaper to transport the cargo by truck or rail; and for how many cities it is cheaper to transport the cargo consolidated or deconsolidated.

The impact of considering an environmental cost in the case of São Paulo is demonstrated by comparing the results of SIM1 and SIM2 with SIM3 and SIM4, where the carbon price makes the kind of transportation change from truck to train in the cheapest TS. Table 2 shows that, for TTD = 2, transport by rail became the cheapest option for 61 cities (from 96 cities in SIM1 and SIM2 to 157 cities in SIM3 and SIM4). For TTD = 5, only seven cities changed to rail for high cargo values (SIM1 compared with SIM3). This included transferring services from seaport to dry ports. However, as the TTD increases, the EMC_{TS} was found to have a very limited impact on the client's choice in terms of cost.

5. Discussion

As relates to addressing our research question (*RQ: How do transport characteristics drive joint seaport–hinterland network decisions in terms of total container shipping cost?*), the outcome of the simulations is that (i) cargo value, (ii) cargo accommodation type, (iii) transportation mode, (iv) delivery distance, and (v) the time until the client receives their cargo impact the seaport–hinterland network decision. The first outcome of the simulation is that, for a short storage time, cargo delivery directly from the seaport is cost-efficient, particularly for low cargo values and for delivery to locations (cities) next to seaports. These findings are aligned with the literature (e.g., Tsao & Linh, 2018). Furthermore, our finding here also suggests that, as storage time increases, use of dry ports becomes the most cost-efficient option. However, as seaports charge a higher storage fee, dry ports appear especially attractive for relatively long-term storage (Kim & Kim, 2007; Qiu et al., 2015).

The second outcome of the simulations is that road transportation is very cost-efficient for most cities in São Paulo state. As dry ports are not directly connected with seaports, an additional charge to transfer the cargo from rail to road is required to use multimodal transportation in the flow of goods. By adding emission cost into the models in SIM3 and SIM4, for a short period of storage, multimodal transportation became the cost-efficient option for 157 cities; particularly for cities far from seaports. This indicates that multimodal transportation may be improved in Brazil, mainly by further connecting dry ports to multimodal facilities, thereby reducing the total logistic costs for clients and achieving the best benefit of scale economies in dry port operations (Khaslavskaya & Roso, 2020). This result is also line with Wang et al. (2020).

Despite the short set of additional services considered in the simulations, the third outcome is that container deconsolidation, and the need to depot empty containers, has impacted the results for joint seaport–hinterland choices. For clients located far from seaports, deconsolidation charges are compensated by avoiding the return of the empty container. Furthermore, the demurrage and detention charges stood out as a driver for client decisions to deconsolidate cargo, mainly for low cargo value.

Last, the average cost of each TS in Fig. 6 has illuminated the difference of total logistics cost by each delivery strategy, indicating benefits for importers in Brazil. Hence, the cost model proved to be a flexible tool to assess the total logistics cost of the seaport-hinterland network, bringing relevant insights for importers/exporters, policy makers, and service providers. Despite the simulations offering clear best route recommendations, each decision maker should, nonetheless, take into account their specificities, logistics capacities, service levels required, and preferences for seaport–hinterland networks that best fit their needs, giving broad regard to factors that go beyond those used for cost analysis. However, as the cost is one of the most important criteria for all logistics decision, the insights from the simulations may also be used in shortcut decision making, particularly under/due to time and information limitations.

6. Conclusion

The primary concern of this study is to inform joint seaport–hinterland network decisions by freight forwarders and clients who directly purchase their services. To that end, we developed a model that can deliver reductions in total costs associated with container shipping. Via simulation, we explored three specific decisions; namely: (i) the decision to perform all logistics services in seaport or in dry port; (ii) the decision to transport the cargo by road or by multimodal transportation (road

and rail); and (iii) to deliver/receive the cargo consolidated or deconsolidated.

Considering seven Transportation Strategies (TS), we found the most cost-efficient strategies as (i) TS1—when the container is directly delivered to the client by truck from the seaport then returns empty to the same seaport; (ii) TS2—when the container is directly delivered to the client by multimodal transport (rail and truck) from the seaport then returns empty through the same route; and (iii) TS3—when the container is deconsolidated in the seaport and delivered directly by truck to the client from the seaport. More specifically, the results from the simulation show that the proposed cost model works in a flexible manner and may be applicable in different countries and contexts once adjusted to client specificities. A finding of the simulation is, therefore, that dry ports are very often cost-efficient. This finding carries the major implication of encouraging Brazilians' shippers and consignees to consider that these additional 'nodes' do not necessarily bring additional operational costs.

Our study makes contributions to both theory and practice. In terms of theoretical contributions, first, our study reiterates within the specific context of Brazil that being able to offer a wide range of service options to a range of clients/stakeholder groups, in a manner that also ensures a reduction in total cost involved in international logistics, entails considering multiple and multiple and complex criteria that bear upon the total costs involved. Second, our study contributes to the literature on joint-choice of services providers to aid seaport–hinterland decisions in container shipping, specifically as relates to decisions made by clients (shippers and consignees) which are focused on reducing the total costs for container shipping among other factors. By setting the context of our study within Brazil (particularly considering our earlier citation of the country's position in the global container-handling economy/ maritime cargo transportation), we have provided a more nuanced and contextualized insight into the critical factors impacting shippers and consignees. In doing so, we have contributed to the extension of research in this area.

Our study also makes contributions to practice. First, the seven TS and 10 cost factors represent useful focal points for critical awareness and a pivotal field of action which can be employed to motivate and inform service providers (e.g., seaports, dry ports, and carriers) and clients on the necessity of collaborating during decision making as a means to ensure successful cost reductions. Such collaboration is required by a need for container shipping stakeholders to recognize that they all operate within a seaport–hinterland eco-system that is dominated by interrelationships. To this effect, our model recognizes the need for a stakeholder-centric strategy towards cost reduction in seaport–hinterland decisions in container shipping. Last, the development of our model is beneficial for both future port research and practice as it allows the conceptualization of container shipping offerings from a cost perspective.

Despite the noted benefits of the model, it does also have some limitations. One limitation is that, despite cost being one of the most important factors in any logistics decision, qualitative criteria must be taken into consideration, which was not the case in the particular instance of the present study. This serves as the foundation for future studies. Furthermore, we encourage researchers to apply the proposed cost model in diverse scenarios and countries, comparing the results and insights highlighted herein; such application is possible by changing some parameters of the model to adjust it for the specific seaport–hinterland network under study.

CRediT authorship contribution statement

Thiago de Almeida Rodrigues: Conceptualization, Methodology, Formal analysis, Writing – original draft. **Caroline Maria de Miranda Mota:** Validation, Formal analysis, Writing – review & editing, Supervision, Funding acquisition. **Udechukwu Ojiako:** Conceptualization, Methodology, Writing – original draft, Supervision. **Maxwell Chipulu:** Formal analysis, Writing – review & editing. **Alasdair Marshall:** Writing – review & editing. **Fikri Dweiri:** Writing – review & editing, Supervision.

Declaration of Competing Interest

None.

Data availability

Data will be made available on request.

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Appendix A. Variables and parameters of the model

Variables and Parameters	Description	Unit of measure	Assumptions	References
Variables				
$ttsc_{TS_t,\ r}$	Transit time from the seaport (i) to the customer's facility (k) by truck (t) or rail (r)	Days	Estimated multiplying the distance and the average speed.	Larranaga et al. (2016).
$ttdc_{TS_t}$	Transit time from the dry port (j) to the customer's facility (k) by truck (t)	Days	Estimated multiplying the distance and the average speed.	Larranaga et al. (2016).
$ttsd_{TS_t,\ r}$	Transit time from the seaport (i) to the dry port (j) by truck (t) or rail (r)	Days	Estimated multiplying the distance and the average speed.	Larranaga et al. (2016).
$ttre_{TS_t,\ r}$	Transit time to return the empty container (empt) by truck (t) or rail (r)	Days	TS = 1 and 2 $ttre_{t,\ r} = ttsc_{t,\ r}$ TS = 4 and 5 $ttre_t = ttdc_t$ TS = 3, 6, and 7 $ttre_{t,\ r} = 0$	Larranaga et al. (2016).
ttd	Time between container discharge at the seaport and its arrival at the customer facility	Days	Varying from 2 days to 60 days.	Defined by the authors
ttc_{TS}	Container transit time by Transportation Strategy (TS)	Days	Operational process of each TS: TS = 1, 2, 4, 5 $ttc_{TS} = TLT_{TS}$ TS = 3 $ttc_{TS} = 0$	Defined by the authors

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Variables and Parameters	Description	Unit of measure	Assumptions	References
			TS = 6, 7 $ttc_{TS} = ttsd_{TS_{i,r}} + dtd_{i,j}$	
$doc_{TS_{i,j}}$	Dwell-time of the cargo in the seaport (i) and dry port (j)	Days	- -	Defined by the authors
$dd_{TS,TM}$	Delivery distance by Transportation Strategy (TS) and Transportation Mode (TM)	Kilometers	Shortest distance by Google Maps.	Google Maps
cv	Cargo value	R$	cv = cifr × cw × dor For simulation 1: R$415,580.00 (US$74,000.00) For simulation 2: R$45,000.00 (US$8040.00)	DP World (2021); Lättilä et al. (2013); Zhang and Schramm (2020).
$box_{i,j}$	Quantity of boxes handled, stored, or delivered through seaport (i) and dry port (j)	Integer (Z)	1 container 40 ft (2 TEUs)	Defined by the authors
$qhm_{TS_{i,j}}$	Quantity of handling moves in seaport (i) and dry port (j) by Transportation Strategy (TS)	Integer (Z)	TS1 : $qhm_i = 1$; $qhm_j = 0$ TS2 : $qhm_i = 1$; $qhm_j = 0$ TS3 : $qhm_i = 0$; $qhm_j = 0$ TS4 : $qhm_i = 1$; $qhm_j = 2$ TS5 : $qhm_i = 1$; $qhm_j = 2$ TS6 : $qhm_i = 1$; $qhm_j = 1$ TS7 : $qhm_i = 1$; $qhm_j = 1$	Defined by the authors
Parameters				
$thc_{i,j}$	Terminal handling charge per handle in seaport (i) and dry port (j)	R$/handle	Average fee in seaports (i) and dry ports (j): thc_i = R$582.00/handle thc_j = R$286.00/handle	Embraport (2021); BTP (2021); Santos Brasil (2021); Deicmar (2021); Eudmarco (2021); Lachmann (2021); Multilog (2021).
scc_{TS}	Storage charge per consolidated box (con) by Transportation Strategy (TS)	% cv/day	Average rate in seaports (i) and dry ports (j): $scc_i = src_i \times cv$ $scc_j = src_j \times cv$	Embraport (2021); BTP (2021); Santos Brasil (2021); Deicmar (2021); Eudmarco (2021); Lachmann (2021); Multilog (2021).
scd_{TS}	Storage charge per deconsolidated cargo (deco) by Transportation Strategy (TS)	% cv/day	Average rate in seaports (i) and dry ports (j): $scd_i = srd_i \times cv$ $scd_j = srd_j \times cv$	Embraport (2021); BTP (2021); Santos Brasil (2021); Deicmar (2021); Eudmarco (2021); Lachmann (2021); Multilog (2021).
cv	Cargo value	R$	cv = cifr × cw × dor For simulation 1: R$415,580.00 (US$74,000.00) For simulation 2: R$45,000.00 (US$8040.00)	DP World (2021); Lättilä et al. (2013); Zhang and Schramm (2020).
cifr	CIF rate	US$/Kg	4,95	Zhang and Schramm (2020).
cw	Cargo weight	Kg	15,000	Lättilä et al. (2013))
dor	Dollar rate	R$ /US$	5,6	Dollar rate in February 27, 2021.
$src_{i,j}$	Storage rate for consolidated cargo in seaport (i) and dry port (j)	Percentage (%)	Average rate in seaports (i) and dry ports (j): src_i = 0.255% src_j = 0.114%	Embraport (2021); BTP (2021); Santos Brasil (2021); Deicmar (2021); Eudmarco (2021); Lachmann (2021); Multilog (2021).
$srd_{i,j}$	Storage rate for deconsolidated cargo in seaport (i) and dry port (j)	Percentage (%)	Average rate in seaports (i) and dry ports (j): srd_i = 0.314% srd_j = 0.137%	Embraport (2021); BTP (2021); Santos Brasil (2021); Deicmar (2021); Eudmarco (2021); Lachmann (2021); Multilog (2021).
$msrc_{i,j}$	Minimum storage rate to consolidated cargo in seaport (i) and dry port (j)	R$	Average rate in seaports (i) and dry ports (j): $msrc_i$ = R$1,888.00 $msrc_j$ = R$2,000.00	Embraport (2021); BTP (2021); Santos Brasil (2021); Deicmar (2021); Eudmarco (2021); Lachmann (2021); Multilog (2021).
$msrd_{i,j}$	Minimum storage charge to deconsolidated cargo in seaport (i) and dry port (j)	R$	Average rate in seaports (i) and dry ports (j): $msrd_i$ = R$2,329.00 $msrd_j$ = R$2,000.00	Embraport (2021); BTP (2021); Santos Brasil (2021); Deicmar (2021); Eudmarco (2021); Lachmann (2021); Multilog (2021).
$dtd_{i,j}$	Documentation time to liberate the cargo at seaport (i) and dry port (j)	Days	2 days for green channel	RFB (2019).
kas_{TS}	Kind of additional services required by the Transportation Strategy (TS)	- -	TS1 = kas_4 is required TS2 = kas_4 is required TS3 = kas_1, kas_2, kas_3, kas_4 is required TS4 = kas_4 is required TS5 = kas_4 is required TS6 = kas_1, kas_2, kas_3, kas_4 is required TS7 = kas_1, kas_2, kas_3, kas_4 is required	Defined by the authors
$cas_{kas_{i,j}}$	Charge per kind of additional service in seaport (i) and dry port (j)	R$/box	Average rate in seaports (i) and dry ports (j): cas_{kas1_i} = R$528.67; cas_{kas1_j} = R$225.71 cas_{kas2_i} = R$1,182.00; cas_{kas2_j} = R$725.46	Embraport (2021); BTP (2021); Santos Brasil (2021); Deicmar (2021); Eudmarco (2021); Lachmann (2021); Multilog (2021).

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Variables and Parameters	Description	Unit of measure	Assumptions	References
			cas_{kas3_i} = R$1,367.67; cas_{kas3_j} = R$725.46 cas_{kas4_i} = R$108.33; cas_{kas4_j} = R$96.17	
$pic_{i,\ j}$	Physical inspection charge in seaport (i) and dry port (j)	R$/box	Average rate in seaports (i) and dry ports (j): pic_i = R$528.67 pic_j = R$199.14	Embraport (2021); BTP (2021); Santos Brasil (2021); Deicmar (2021); Eudmarco (2021); Lachmann (2021); Multilog (2021).
$xsc_{i,\ j}$	X-ray scanner charge in seaport (i) and dry port (j)	R$/box	Average rate in seaports (i) and dry ports (j): xsc_i = R$462.67 xsc_j = R$333.00	Embraport (2021); BTP (2021); Santos Brasil (2021); Deicmar (2021); Eudmarco (2021); Lachmann (2021); Multilog (2021).
$tcc_{i,\ j}$	Transshipment customs charge to transfer the cargo from seaport (i) to dry port (j)	R$/box	Average rate in seaports (i) and dry ports (j): tcc_i = R$1,323.36 tcc_j = R$176.00	Embraport (2021); BTP (2021); Santos Brasil (2021); Deicmar (2021); Eudmarco (2021); Lachmann (2021); Multilog (2021).
$as_{t,\ r}$	Average speed of truck (t) and rail (r)	Km/h	as_t = 60km/hr as_r = 40 km/hr	Larranaga et al. (2016).
or	Opportunity cost rate	Percentage (%)	or + der = 0.0685%	Tran et al. (2017).
der	Depreciation cost rate	Percentage (%)	or + der = 0.0685%	Tran et al. (2017).
ddft	Demurrage and detention free-time	Days	Average rate among ship-owners: ddft = 7	MSC (2021); Maersk (2021); CMA CGM (2021); Hapag-Lloyd (2021); Cosco (2021).
ddf1	Demurrage and detention fee 1	R$/day	Average rate among ship-owners: ddf1 = 72.67	MSC (2021); Maersk (2021); CMA CGM (2021); Hapag-Lloyd (2021); Cosco (2021).
ddf2	Demurrage and detention fee 2	R$/day	Average rate among ship-owners: ddf2 = 127.17	MSC (2021); Maersk (2021); CMA CGM (2021); Hapag-Lloyd (2021); Cosco (2021).
lcd	Leasing charge per day	R$/day	lcd = 8.96	Tran et al. (2017))
$mc_{t_{con,\ deco,\ empt},\ r_{con,\ empt}}$	Modal charge by Transportation Mode (TM) and Cargo Accommodation Type (CT)	R$/box*Km	Average rate among carriers: $mc_{t_{con}}$ = R$4.114 $mc_{t_{deco}}$ = R$4.114 $mc_{t_{empt}}$ = R$3.773 $mc_{r_{con}}$ = R$3.591 $mc_{r_{empt}}$ = R$1.716	ANTT (2021); Araújo Maria da Penha et al. (2014); NTC (2021); Tran et al. (2017).
$fdr_{t_{con,\ deco,\ empt},\ r_{con,\ empt}}$	Fixed delivery rate by Transportation Mode (TM) and Cargo Accommodation Type (CT)	R$	Average rate among carriers: $fdr_{t_{con}}$ = R$785.51 $fdr_{t_{deco}}$ = R$785.51 $fdr_{t_{empt}}$ = R$722.70 $fdr_{r_{con}}$ = R$609.24 $fdr_{r_{empt}}$ = R$437.00	ANTT (2021); Araújo Maria da Penha et al. (2014); NTC (2021).
adc_{TS}	Additional delivery cost for transportation under customs control by Transportation Strategy (TS)	R$	For TS = 4, 5, 6, and 7: $adc_{t_{con},\ r_{con}}$ = R$924.57	NTC (2021).
otr_{TS}	Operational transshipment rate by Transportation Strategy (TS)	R$	For TS = 2, 4, and 7: otr_{TS} = R$276.77	ANTT (2021).
cr	Carbon rate	R$/CO2tonn	*cr = R$280.00/CO2ton*	EDF (2020))
$me_{t,\ r}$	Marginal emission	KgCO2/tonn. km	me_t = 0.044 KgCO2/t. km me_r = 0.0067 KgCO2/t. km	Lättilä et al. (2013).
$w_{t,\ r}$	Weight of the cargo plus the truck (t) or wagon (r)	Tones	w_t = 30 t when cw = 15 t w_r = 47.76 t when cw = 15 t	Lättilä et al. (2013).

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